

Weekly Progress Report

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Domain: Internet of Things (IoT) / Electronics
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I. Overview

This week, the primary focus was on analyzing and conceptualizing an IoT-based Air Quality Monitoring System utilizing the UCT Data Acquisition and Monitoring System (DAMS) framework.

II. Achievements

1. USC_TIA Familiarization:

- Evaluated how UCT DAMS aggregates real-time environmental data streams from gas and particulate sensors.

2. Project Contributions:

- **Name of the project:** IoT-based Air Quality Monitoring System
- Mapped out the sensor integration layout for detecting hazardous gases, CO2 levels, and particulate matter (PM2.5 / PM10).
- Designed the logical threshold limits within the monitoring system to trigger automated alerts when air quality indices drop below safe levels.

3. Learning Python & IoT Programming:

- Studied data visualization techniques in Python to plot air pollution trends over specified time intervals.

III. Challenges

1. USC_TIA Integration:

- Encountered calibration complexities when interfacing heterogeneous electrochemical gas sensors with the standard DAMS input channels.

2. Project Complexity:

- Faced difficulties handling erratic or noisy sensor fluctuations caused by rapid outdoor ambient temperature shifts.

IV. Learning Resources

1. USC_TIA Documentation:

- Referenced the UCT DAMS product manuals regarding multi-channel sensor logging and external transceiver configurations.

2. IoT Learning Resources:

- Studied international Air Quality Index (AQI) math models to implement correct scaling algorithms in the software logic.

V. Next Week's Goals

1. USC_TIA Enhancement:

- Set up a hardware simulation to test data packet transmission stability from the sensor node to the DAMS cloud interface.

2. Project Development:

- Write a Python script to filter out random sensor spikes using moving average noise reduction.

VI. Additional Comments

Transitioning the DAMS architecture toward an environmental health application highlights the framework's scalability. Overcoming sensor calibration challenges this week has deepened my understanding of practical hardware deployment variables.